Pedro Brunialti Lima de Andrade

Office; 119 Kerchof Hall, 141 Cabell Drive, Charlottesville, VA 22904, USA
eyu4dh@virginia.edu pbrunialti.github.io/

Areas of interest

Homotopy Theory, Algebraic Topology, and Mathematical Physics

Education

2022-Present: **Title: Involutions on (∞, n)-categories of bordisms and reflection TQFTs**, *University of Virginia*, Charlottesville. Ph.D. in Mathematics – Advisor: Julie Bergner, expected May 2027

2020-2022: **Title: Comparing Models for $(\infty, 1)$-categories.**, *Federal University of Minas Gerais - UFMG*, Belo Horizonte. M.Sc. in Mathematics – Advisor: Rodney Josué Biezuner

2016-2019: **B.Sc. in Physics**, *Federal University of Minas Gerais - UFMG*, Belo Horizonte

Papers and preprints

[1] Abdelmalek Abdesselam, Pedro Brunialti, Tristan Doan, and Philip Velie. A bijection for tuples of commuting permutations and a log-concavity conjecture. *Res. Number Theory*, 10(2):Paper No. 45, 10, 2024.

[2] Julie Bergner, Pedro Brunialti, Eleftherios Chatzitheodoridis, and Josh Turner. Trees associated with unitary partition complexes, in preparation.

[3] Anna Marie Bohmann, Pedro Brunialti, Bertrand Guillou, David Mehrle, and Chris Portwood. Free bi-incomplete Tambara functors are almost never free, in preparation.

[4] Pedro Brunialti. Involutions on (∞, n)-categories of bordisms and reflection TQFTs. 2026. Preprint, https://arxiv.org/abs/2610.23712.

Invited talks

04/2027 (planned): **Free bi-incomplete Tambara functors are almost never flat**, AMS Eastern Sectional. Special Session on Homotopy Theory and Algebraic K-Theory , Syracuse, New York

01/2027 (planned): **Involutions on (∞, n)-categories of bordisms and reflection TQFTs**, Joint Mathematics Meeting. AIM Special Session on (∞, ∞)-Categories, Chicago, Illinois

12/2026 (planned): **Involutions on (∞, n)-categories of bordisms and reflection TQFTs**, Canadian Mathematical Society Meeting. Special Session on Higher Category Theory Across Mathematics, Montréal, Québec

10/2026: **Involutions on (∞, n)-categories of bordisms and reflection TQFTs**, Geometry and Topology Seminar. Virginia Commonwealth University, Richmond

04/2026: **Free bi-incomplete Tambara functors are almost never flat**, 5 College Geometry-Topology Seminar. University of Massachusetts, Amherst

Contributed talks

10/2026: **Involutions on (∞, n)-categories of bordisms and reflection TQFTs** , 2026 Fall Eastern Sectional Meeting. George Washington University, DC

09/2026: **Involutions on (∞, n)-categories of bordisms and reflection TQFTs** , Union College Mathematics Conference. Union College, New York

06/2026: **Walker–Wang models and braided categorical nets**, Talbot Workshop. Rangeley, Maine

05/2026: **(∞, n)-symmetric monoidal categories with involutions**, Higher Differential Geometry. University of Greifswald, Germany

02/2026: **(∞, n)-symmetric monoidal categories with involutions**, UVA Topology Seminar

11/2025: $(\infty, 2)$**-symmetric monoidal categories with involutions**, BUGCAT Conference. Binghamton University, New York
09/2025: **Chern classes in physics**, UVA Graduate Seminar
08/2024: **Understanding and classifying TQFTs**, UVA Graduate Seminar
01/2024: **What are universal properties?**, UVA Graduate Seminar
01/2023: **A mathematical formulation of quantum mechanics**, UVA Graduate Seminar
09/2021: **Equivalence between homotopy theories**, I Mathematics Meeting of Minas Gerais
01/2021: **Modern homotopy theory: model categories**, First Brazilian Meeting on Category Theory
01/2021: **Tutorial: categories and functors**, First Brazilian Meeting on Category Theory
06/2021: **Interpreting the measurement axiom**, Group for Fundamental Theory (UFMG)
06/2021: **The mathematics of gauge theories**, Group for Fundamental Theory (UFMG)
01/2021: **The Serre-Swan Theorem and K-theory**, Math-Phys-Cat group
10/2020: **An invitation to model categories**, Math-Phys-Cat Group

Teaching experience

Fall 2026: **Instructor of record**, *Calculus II*, Mathematics Department, University of Virginia
Fall 2025: **Instructor of record**, *Calculus II*, Mathematics Department, University of Virginia
Fall 2024: **Instructor of record**, *A Survey of Calculus I*, Mathematics Department, University of Virginia
Spring 2024: **Instructor of record**, *Calculus I*, Mathematics Department, University of Virginia
Fall 2023: **Instructor of record**, *A Survey of Calculus II*, Mathematics Department, University of Virginia
Spring 2023: **Teaching assistant**, *Calculus II*, Mathematics Department, University of Virginia
Fall 2022: **Graduate teaching assistant**, *Calculus III*, Mathematics Department, University of Virginia
10/2021-01/2022: **Graduate teaching assistant**, *Differential Equations A* , Dep. of Mathematics, ICEx, UFMG
05/2021-09/2021: **Graduate teaching assistant**, *Analytic Geometry and Linear Algebra*, Dep. of Mathematics, ICEx, UFMG

Mentoring experience

University of Virginia Math Experimental Lab
Spring 2026: Co-mentor: Thomas Jaklitsch. Project: Compass and Straightedge Constructions.

University of Virginia Directed Reading Program
Fall 2024: Mentee: Zubin Kaul. Project: Differential geometry of curves and surfaces.
Spring 2024: Mentee: Mary Edith Plunkett. Project: Nonlinear dynamics and chaos.
Fall 2023: Mentee: Brian Meiser. Project: Hamiltonian mechanics and Poisson brackets.

Conferences, schools, and workshops attended

Northeast Conference on Categorical Methods, Harvard University *2026*
2026 Fall Eastern Sectional Meeting, Washington, DC, USA *2026*
Union College Mathematics Conference, Schenectady, New York, USA *2026*
International Congress of Mathematicians, Philadelphia, Pennsylvania, USA *2026*
DV60: Discoveries and Vistas, Johns Hopkins University *2026*
Talbot Workshop 2026, Rangeley, Maine, USA *2026*
Atlantic TQFT Spring School 2026, Dalhousie University *2026*
BTW6: The Volume Conjecture, Frejus, France *2026*

Hgher Differential Geometry, Greifswald, Germany *2026*

GSTGC 2026, Rutgers University University *2026*

BUGCAT Conference, Binghamton University *2025*

Mathematical Congress of the Americas 2025, Miami, Florida, USA *2025*

Mid-Atlantic Collaborative Workshop in Algebraic Topology, Lexington, Virginia, USA *2025*

Atlantic TQFT Spring School 2025, University of New Brunswick *2025*

AMTRaK Conference, University of Pennsylvania *2025*

Mid-Atlantic Topology Conference, University of Virginia *2025*

New England Algebraic Topology and Mathematical Physics, Providence College *2024*

Encuentro Colombiano de Geometría y Topología, National University of Colombia *2024*

Atlantic TQFT Spring School 2024, Memorial University Newfoundland *2024*

Graduate Student Topology and Geometry Conference, Michigan State University *2024*

BUGCAT Conference, Binghamton University *2023*

Algebraic K-Theory and Redshift, Johannes Gutenberg University Mainz *2023*

Topological Moonshine, University of Illinois, Urbana-Champaign *2023*

Atlantic TQFT Spring School 2023, Wolfville, Nova Scotia, Canada *2023*

Homotopy Theory in Honor of Paul Goerss, Northwestern University *2023*

Field of Dreams Conference, Minneapolis, Minnesota, USA *2022*

I Mathematics Meeting of Minas Gerais, virtual participation due to COVID-19 *2021*

First Brazilian Meeting on Category Theory, virtual participation due to COVID-19 *2021*

Toposes online, virtual participation due to COVID-19 *2021*

Scholarships

2020-2022: **Academic Excellence Program (PROEX)**. M.Sc. program scholarship.
2018-2019: **Program of Undergraduate Research Scholarships (CNPq)**. Undergraduate research scholarship.

Academic service

Fall 2026 - Present: **Member of the directed reading program (DRP) committee**
Fall 2024 - Spring 2026: **President of the University of Virginia AMS Graduate Student Chapter**

Languages

Portuguese: Native
English: Fluent
Spanish: Advanced
French: Intermediate